

Sanjith S J

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EDUCATION

Amrita Vishwa Vidyapeetham

B.Tech in Computer Science CGPA: 8.8/10

Coimbatore, TN

Jul 2024 – May 2028

- **Relevant Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks

EXPERIENCE

Undergraduate Teaching Assistant

Amrita School of Computing

Sep 2025 – Present

Coimbatore, TN

- Designed and deployed a Python-based autograding system using PyTest, reducing grading turnaround from **~2 weeks to <1 hour** for 500+ students.
- Built a scalable full-stack TA management portal using Next.js and Supabase to streamline application review and onboarding for **50+ candidates per semester**.
- Analyzed Git commit histories across 15+ student teams to detect collaboration anti-patterns; **proposed and helped implement** revised project evaluation protocols.
- Engineered automated reporting pipelines that improved grading consistency and eliminated human data-entry errors at scale.

PROJECTS

SaaS Event Synchronization Engine

Python, Next.js, PostgreSQL, AWS Lambda, OAuth

Jun 2025 – Present

- Architected an event-driven SaaS platform that synchronizes high-priority schedules between WhatsApp, Google Calendar, and Notion in real-time.
- Engineered a fault-tolerant ingestion pipeline using **AWS Lambda** and webhooks to process asynchronous payloads with 99.9% reliability.
- Implemented secure OAuth 2.0 authentication flows to manage user sessions and API tokens across multiple third-party services.

High-Performance Skip List Engine

Java, Spring Boot, Maven, D3.js

Jan 2025 – Feb 2025

- Implemented a custom Skip List data structure from first principles in Java, achieving $O(\log n)$ average-time search and insertion.
- Developed an interactive D3.js visualization to demonstrate probabilistic balancing behavior on datasets exceeding **10k nodes**.
- Exposed RESTful APIs via Spring Boot to dynamically transform linear data structures into optimized Skip List representations.

Embedded Motorcycle Telemetry System

C++, Arduino, I2C, Accelerometers

Sep 2025 – Oct 2025

- Developed a "BlackBox" accident detection system using **C++** and Arduino, processing real-time 6-axis IMU sensor data to detect crash signatures.
- Implemented memory-efficient circular buffers to log and retain the final 30 seconds of telemetry data prior to impact events.
- Optimized sensor polling algorithms to maintain high-frequency data capture within the constraints of limited-resource microcontrollers.

TECHNICAL SKILLS

Languages: Java, Python, C++, SQL, JavaScript

Backend: Spring Boot, Flask, RESTful APIs, OAuth, AWS Lambda

Frontend: React, Next.js, Tailwind CSS, D3.js

Systems & Tools: Git, Docker, AWS, GCP, Linux, PostgreSQL, TimescaleDB